Lecture 1: Real Numbers

AMA3707 Real Analysis

1 Numbers

Definition 1.1 (Natural Numbers)**.** The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

Convention 1.1 (Natural Numbers Exclude Zero)**.** In these notes, $0 \notin \mathbb{N}$.

Definition 1.2 (Integers)**.** The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

Definition 1.3 (Rational Numbers)**.** A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}.$$

Remark 1.1. Under the usual identifications, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$.

2 Fields

Definition 2.1 (Closed Operation)**.** An operation $*$ is said to be *closed* on a set A if for all $a, b \in A$, we have $a * b \in A$.

Definition 2.2 (Field)**.** A *field* is a set F equipped with addition and multiplication such that both operations are multiplication are

- closed operations on F,
- associative, meaning for all $a, b, c \in F$, we have $(a+b)+c = a+(b+c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$,
- commutative, meaning for all $a, b \in F$, we have $a + b = b + a$ and $a \cdot b = b \cdot a$,
- have identity elements 0 and 1,
- each element has an additive inverse, such that for each $a \in F$, there exists $-a \in F$ with $a + (-a) = 0$,
- each non-zero element has a multiplicative inverse, such that for each $a \in F$ with $a \neq 0$, there exists $a^{-1} \in F$ with $a \cdot a^{-1} = 1$

Proposition 2.1 ($\mathbb{Q}$ Is a Field)**.** *The rational numbers with their usual addition and multiplication form a field.*